

YAX Major-Revision Diagnosis

Evidence-led framing decision and remaining work

September 2026

Decision

The paper should be a **composition-centered measurement paper**, titled *AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment*. The strongest defensible substantive result is that the public-CPS young-relative association is largely a comparison across broad occupation families. The strongest methodological result is that implementation repair, same-construct measurement, construct choice, support, estimand, representation, and coordinate changes have to be audited separately.

The earlier thesis—that robustness is generally statement-specific—was true but too generic. The stronger revised thesis is empirical and falsifiable: after correcting the data, a $-.1346$ Q5–Q1 association falls to about $-.032$ when broad-family evolution is absorbed, while the design is too imprecise to resolve most paired architecture differences. The chapter is not a causal AI-displacement paper.

Evidence that forced the decision

1. **Broad-family conditioning.** SOC2-by-post and SOC2-by-month specifications yield $-.0315$ and $-.0317$, versus $-.1346$ at baseline. Paired movements of approximately $.103$ have intervals excluding zero.
2. **Support cost.** Only four of 22 SOC2 families contain both Q1 and Q5, and conditioned target information is 30 percent of baseline. The residual detailed-exposure coefficient is weakly measured and is not a causal remainder.
3. **Food-service mechanism rejected.** Removing Q1 food-service occupations attenuates the coefficient only 8.8 percent. Broader in-person-service exclusions make it slightly more negative.
4. **Architecture heterogeneity unresolved.** The primary MDE_{80} is $.1244$. All seven paired beta-minus-alternative Q5–Q1 intervals include zero; paired MDEs reach $.1689$. Point-estimate differences cannot carry claims of detected heterogeneity or equivalence.
5. **Direct public BCC bridge.** Their top-two-versus-bottom-three beta grouping produces a smaller $-.0728$ CPS coefficient. This anchors the audit to the motivating classification while preserving the boundary around proprietary hiring evidence.
6. **Construction dependence.** Two components explain 96.11 percent of the six core score

implementations. The prior six-measure sign exercise is a dependent-set statement, not six independent validations.

7. **Data corrections do not explain the negative association.** March repair moves $-.1311$ to $-.1346$. One-to-one-target and stable-taxonomy estimates remain negative. Population-control and respondent-equivalent diagnostics do not reveal a mechanical January-2025 explanation.
8. **Mobility is diagnostic rather than mechanistic.** A family-balanced pairwise statistic is 20.13 percent, while 53.28 percent is a six-way “any conflict” event. The aligned-support realized-minus-rematching gap is 0.959 point with a declared household-cluster sampling SE of 0.0776 point, but the algorithm-defined benchmark and switch-only sample do not identify hiring or coding error.

What the revised paper may say

It may say that public CPS data reproduce a negative young-relative association under beta exposure and the public BCC grouping; that broad-family conditioning absorbs most of the point-estimate magnitude; that the movement is detectable but changes the estimand; that food-service exclusions do not explain the result; that mapping errors can remove nearly all computer/math coverage; and that available precision does not resolve most architecture differences.

It may not say that AI caused employment loss, that 76 percent is a causal share, that the remaining $-.032$ is an AI effect, that food-service recovery is the mechanism, that intervals containing zero establish equivalence, that six measures independently validate the finding, or that CPS reproduces ADP hiring and separation evidence.

Major-comment completion

Both reports’ numbered major comments have been translated into an exact response matrix. The major empirical program now covers composition, profile and monotonicity, stable tails, joint influence, service exclusions, canonical intervals, realized MDEs, quarterly aggregation, time-score covariance, pre-AI pseudo-breaks, population controls, the BCC grouping, architecture dependence, corrected mapping, age-allocation uncertainty, and mobility/rematching uncertainty. The manuscript and appendix have been reorganized around the economic question rather than revision chronology.

Two requests receive principled non-adoption. No counterfactual January-2025 age-by-occupation weights are created because official subgroup bridge factors do not exist. No national adoption or industry BTOS series is appended merely to reparameterize event time or introduce an unvalidated occupation–industry bridge. Both limits are explicit.

Remaining work before submission

This is the **major-advice revision**, not the final copyedit. The next pass should reconcile all secondary comments line by line, finalize citations and table formatting, and ask the author to complete affiliation, email, acknowledgments, funding, conflicts, and data-use disclosures. The paper should also receive an independent econometric review of the maintained occupation-shock inference before being sent to a top journal. These open items do not alter the framing decision.